

PRANJAL KATTE

Indian Institute of Technology, Madras

in [Pranjal Katte](#) @ pranjalkatte82@gmail.com ☎ +91 9850951046



EDUCATION

B. Tech. Computer Science and Engineering	CGPA : 8.04
Indian Institute of Technology Madras	📅 2022 - 26
X ICSE	98 %
Saint Thomas School	📅 2019
XII HSC	86.17 %
SES Jr.College	📅 2021

WORK EXPERIENCE

Software Engineering Intern - ORACLE INDIA	Summer Internship
Pre-Placement Offer	📅 June '25
Oracle Cloud Team	
- Automated Change Management ticket creation for Oracle IRC deployments using Python, integrating Oracle DevOps Shepherd API, CLI commands, and JSON data. - Cut manual effort by 80 % and reduced ticket creation time by 25+ minutes through end-to-end automation of multi-phase rollouts and data extraction. - Tools : Python, REST APIs, Oracle Shepherd, JSON, JAVA	

CODING PROJECTS

Memory Management System with Low-Level Systems Programming

- Developed a C++ memory management simulator implementing paging, swapping, LRU, and FIFO page replacement algorithms to model real-world operating system memory management scenarios.

Scalar Pipelined Processor Design

- Developed an intricate scalar pipelined processor simulator in C++, tackling the complexities of a five-stage pipeline and handling diverse RISC instructions at the software level.
- Designed efficient cache systems and implemented hazard detection mechanisms to ensure accurate pipeline performance

Deep Learning Model Optimization and Normalisation

- Developed and benchmarked 2 deep learning image-classification systems and MLFFNN/DFNN architectures to evaluate optimization algorithms, achieved convergence in minimum epochs, improved model stability and generalization using Adam and RMSProp optimizers

Academic Entity Management System

- Designed a robust C++ system to efficiently manage and query information for academic entities using dynamic memory allocation, file handling, and object-oriented programming principles.

PacketPulse: PCAP Traffic and HTTP Payload Analyzer

- Developed a network traffic analysis system to process large-scale PCAP traces, analyze throughput for 5 concurrent clients over a 100-second capture window, and automate per-client bandwidth visualization and traffic distribution analysis.

POSITIONS OF RESPONSIBILITY

Manager- Events & Networking

- E-CELL, IIT Madras 📅 Jun '23 - May '24
- Framed and successfully conducted the National Innovation Entrepreneurship Olympiad as a part of E-21- an initiative to empower entrepreneurship in the 21st century students.(1200+ participants)

ACHIEVEMENTS

- Secured **AIR 653** in the JEE-ADVANCED 2022 (out of 1.5Lakh+ candidates)
- Secured **AIR 1723** in the JEE-MAINS 2022 (out of 10Lakh+ candidates)
- Secured **99.95** percentile in MHT-CET 2022 (out of 2.8Lakh+ candidates)
- Secured **AIR 2227** under General category in the Kishore Vaigyanik Protsahan Yojana Exam 2022
- District Topper in Class 10th ICSE Board Exam and JEE ADVANCED 2022
- Gold Medalist** in National Level ABACUS Competition
- Achieved top **20%** global rank in WorldQuant BRAIN's International Quant Championship 2025

RELEVANT COURSES

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| • Probability, Statistics and Stochastic Process | • Object-Oriented Algo Implementation and Analysis |
| • Programming and Data Structures | • Network Security |
| • Design and Analysis of Algorithms | • Computer Organization and Architecture |
| • Basic Graph Theory | • Paradigms of Programming |
| • Operating System | • Languages, Machines and Computation |
| • Discrete Mathematics for CS | • Artificial Intelligence |
| • Compiler Design | • Cloud Computing |
| • Network Security | • ML Foundations |
| • Cloud Computing | • Python for Trading |

CODING EXPERIENCE

- Solved **450+** DSA problems across various coding platforms such as GeeksforGeeks, Codeforces, LeetCode, Codechef, HackerRank
- Max Rating **1669** on CodeChef and Max.Rating of **1494** on Codeforces
- Global Rank **132** in Codechef Starters 165 Contest
- Languages : C, C++, Python, Java, Verilog, HTML, Node JS
- Tools : OCaml, Lex-Yacc, VS Code, $\text{\LaTeX}$, AutoCAD, Git, GitHub, Kubernetes, Docker, Hadoop
- Reached **Gold** level in the World Quant Challenge, designed quantitative alphas and strategies using historical market data

EXTRA CURRICULAR

- Ranked in the **top 10** out of **120** teams in the competitive **Guesstimate Premier League** by TechSoc & Case Club, IIT Madras, showcasing superior analytical and problem-solving expertise
- Representative of the National Cadet Corps (NCC) contingent of IITM (Independence Day Parade Awardee)